

NutriFit

A Simple Health & Nutrition Calculator

Project Report

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1. Introduction

A lot of people want to keep an eye on their health but don't really know basic things like their BMI or how many calories they burn in a day. There are calculators for this online but they are all on different websites, so I decided to make a small Python program called **NutriFit** that puts four common health calculations together in one place: BMI, BMR, TDEE and Body Fat Percentage. It runs in the terminal and shows a menu so the user can pick whichever calculation they want, one after another, without closing and reopening the program every time.

2. Problem Statement

Most people don't have one simple place where they can check their BMI, BMR, TDEE and Body Fat Percentage together – they have to search for each one separately. Also, a lot of online calculators just show a number without telling the user what it actually means, like whether their BMI is in the healthy range or not. This project tries to fix that by making one small program that does all four calculations and also explains the BMI result in simple words (underweight, healthy, overweight or obese).

3. Functional Requirements

The program has a main menu with four calculators. Each one is its own module:

- **Module 1 – BMI Calculator:** User enters height (in metres) and weight (in kg). The program calculates $BMI = \text{weight} / \text{height}^2$ and tells the user if they are underweight, healthy weight, overweight or obese based on the value.
- **Module 2 – BMR Calculator:** User enters gender, height (cm), weight (kg) and age. The program calculates Basal Metabolic Rate using a standard formula (different for male and female).
- **Module 3 – TDEE Calculator:** Same as BMR but the user also picks an activity level (Sedentary, Lightly Active, Moderately Active, Very Active, Extra Active) and the BMR is multiplied by a number depending on the activity level to get the total calories needed per day.
- **Module 4 – Body Fat Percentage Calculator:** User picks a category (Adult male, Adult female, Boy, Girl) and enters height, weight and age. The program calculates an approximate body fat percentage using BMI and age.

How it works: the program keeps showing the menu in a loop. The user types a number (1 to 4) to pick a calculator, or 0 to exit. After a calculation is done, the menu is shown again so the user can do another one if they want.

4. Non-Functional Requirements

- **Usability:** The menu is simple text with numbers, so it's easy to use even for someone who isn't very technical.
- **Maintainability:** The BMI and BMR formulas are written as separate functions (`bmi()`, `bmr_for_male()`, `bmr_for_female()`) instead of writing the formula again and again, so it's easier to fix or change later.

- **Reliability:** The menu loop keeps running until the user chooses to exit (option 0), and if the user types a wrong option it just shows a message and asks again instead of stopping.
- **Resource usage:** The program only does simple maths on numbers the user types, so it runs fast and doesn't need much memory.

5. System Architecture

The program is a single Python file with a simple structure: the top part has the calculation functions (bmi, bmr_for_male, bmr_for_female), and the bottom part has a loop that shows the menu, takes the user's choice, and calls the right function for that choice. There is no database, no file saving and no internet connection used – everything happens while the program is running and is lost once it's closed.

Part of Program	What it does	Where in the code
Menu / Input-Output	Shows the menu, takes input from user, prints results	print() and input() lines
Main loop	Decides which calculator to run based on user's choice	while True: with if/elif
Calculation functions	Actually does the maths for BMI and BMR	bmi(), bmr_for_male(), bmr_for_female()

6. Design Diagrams

6.1 Workflow Diagram (steps)

(written out step by step below — draw this as an actual flowchart image before final submission)

- Start → show main menu
- User enters a choice
- If choice is 1 → run BMI calculator → show result → go back to menu
- If choice is 2 → run BMR calculator → show result → go back to menu
- If choice is 3 → run BMR then TDEE calculator → show result → go back to menu
- If choice is 4 → run Body Fat calculator → show result → go back to menu
- If choice is 0 → print thank you message → program ends
- If choice is anything else → show "Enter a valid choice" → go back to menu

6.2 Use Case Diagram (in words)

User is the only actor. The things the user can do are: Calculate BMI, Calculate BMR, Calculate TDEE, Calculate Body Fat Percentage, and Exit the program. All four calculators can be reached straight from the main menu.

6.3 How the Functions Connect

- Main menu calls → bmi() when the user picks option 1
- Main menu calls → bmr_for_male() or bmr_for_female() when the user picks option 2
- For TDEE (option 3), the program first calls bmr_for_male()/bmr_for_female(), then multiplies the result by an activity number

- For Body Fat (option 4), the program first calls bmi(), then uses that value in the body fat formula

Since the project doesn't use a database or save data to a file, there is no ER diagram in this version.

7. Design Decisions & Rationale

- I made the BMI and BMR calculations into separate functions because they are used in more than one place – BMR is needed for both the BMR option and the TDEE option, and BMI is needed for both the BMI option and the Body Fat option. This way I don't have to write the same formula twice.
- I used a while loop for the menu so the user can do more than one calculation without restarting the program every time, which felt more convenient.
- I kept the program as a simple command-line tool instead of making a GUI, mainly to keep it simple and focus on getting the calculations right first.

8. Implementation Details

Language / tools used: Python 3, only using built-in input()/print() – no external libraries needed.

Formulas used in the code:

- $BMI = \text{weight (kg)} / \text{height (m)}^2$
- $BMR (\text{male}) = 10 \times \text{weight} + 6.25 \times \text{height} - 5 \times \text{age} + 5$
- $BMR (\text{female}) = 10 \times \text{weight} + 6.25 \times \text{height} - 5 \times \text{age} - 161$
- $TDEE = BMR \times \text{activity number (1.2 to 1.9 depending on activity level)}$
- $\text{Body Fat \% (Adult male)} = 1.2 \times BMI + 0.23 \times \text{age} - 16.2$
- $\text{Body Fat \% (Adult female)} = 1.2 \times BMI + 0.23 \times \text{age} - 5.2$
- $\text{Body Fat \% (Boy)} = 1.51 \times BMI + 0.70 \times \text{age} - 2.2$
- $\text{Body Fat \% (Girl)} = 1.51 \times BMI + 0.70 \times \text{age} - 1.4$

The two main functions, straight from the code:

```
def bmi(a, b):
    c=b/(a*a)
    return(c)

def bmr_for_male(w, h, a):
    bmr=(10*w)+(6.25*h)-(5*a)+5
    return(bmr)
```

How the menu works: the whole program runs inside a while True loop. It prints the menu, reads the choice the user types, and uses if/elif to figure out which block of code to run. Typing 0 breaks the loop and ends the program.

9. Screenshots / Results

Below are screenshots of the NutriFit program in Python IDLE: first the code (Figures 1 and 2), and then a sample run of the program (Figure 3).

```
NutriFit.py - C:/Users/Shivam Singh/Documents/NutriFit.py (3.14.7)
File Edit Format Run Options Window Help

def bmi(a,b):
    c=b/(a*a)
    return(c)
def bmr_for_male(w,h,a):
    bmr=(10*w)+(6.25*h)-(5*a)+5
    return(bmr)
def bmr_for_female(w,h,a):
    bmr=(10*w)+(6.25*h)-(5*a)-161
    return(bmr)
print("=====\\n    NutriFit: An Algorithmic Health & Nutrition Analytics Engine\\n    Analytics Engi
while True:
    print("=====\\n1. BMI(Body Mass Calculator)\\n2. BMR(Basal Metabolic Rate)\\n3. TDEE(Total Daily
    choice=int(input("Enter a choice:"))
    if choice==1:
        height=float(input("Enter height in metre:"))
        weight=float(input("Enter weight:"))
        d=bmi(height,weight)
        if d<=18.5:
            print("BMI is",d,"kg/m^2 and the person is underweight")
        elif d<=24.9:
            print("BMI is",d,"kg/m^2 and the person is healthy weight")
        elif d<=29.9:
            print("BMI is",d,"kg/m^2 and the person is overweight")
        else:
            print("BMI is",d,"kg/m^2 and the person is Obese")
    elif choice==2:
        g=input("Person is male or female:")
        hieght=float(input("Enter hieght:"))
        weight=float(input("Enter wieght:"))
        age=int(input("Enter age "))
        if g=="male":
            c=bmr_for_male(weight,hieght,age)
            print("basal metabolic rate is",c,"kilocalories per day")
        if g=="female":
            c=bmr_for_female(weight,hieght,age)
            print("basal metabolic rate is",c,"kilocalories per day")
    elif choice==3:
        gender=input("Person is male or female")
        hieght=float(input("Enter hieght:"))
        weight=float(input("Enter wieght:"))
        age=int(input("Enter age "))
        if gender=="male":
            c=bmr_for_male(weight,hieght,age)
            print("basal metabolic rate is",c,"kilocalories per day")
        elif gender=="female":
            c=bmr_for_female(weight,hieght,age)
            print("basal metabolic rate is",c,"kilocalories per day")
        activity_level=input("Activity level is Sedentary or Lightly Active or Moderately Active or Very Active or Extra Active:")
        if activity_level=="Sedentary":
            tdee=c*1.2
```

Figure 1: NutriFit code, part 1 (functions, main menu, BMI, BMR and start of TDEE)

```
NutriFit.py - C:/Users/Shivam Singh/Documents/NutriFit.py (3.14.7)
File Edit Format Run Options Window Help

weight=float(input("Enter wieght:"))
age=int(input("Enter age "))
if gender=="male":
    c=bmr_for_male(weight,hieght,age)
    print("basal metabolic rate is",c,"kilocalories per day")
elif gender=="female":
    c=bmr_for_female(weight,hieght,age)
    print("basal metabolic rate is",c,"kilocalories per day")
activity_level=input("Activity level is Sedentary or Lightly Active or Moderately Active or Very Active or Extra Active:")
if activity_level=="Sedentary":
    tdee=c*1.2
    print("Total Daily Energy Expenditure of Body is",tdee,"kilocalories per day")
elif activity_level=="Lightly Active":
    tdee=c*1.375
    print("Total Daily Energy Expenditure of Body is",tdee,"kilocalories per day")
elif activity_level=="Moderately Active":
    tdee=c*1.55
    print("Total Daily Energy Expenditure of Body is",tdee,"kilocalories per day")
elif activity_level=="Very Active":
    tdee=c*1.725
    print("Total Daily Energy Expenditure of Body is",tdee,"kilocalories per day")
elif activity_level=="Extra Active":
    tdee=c*1.9
    print("Total Daily Energy Expenditure of Body is",tdee,"kilocalories per day")
else:
    print("Enter a valid input")
elif choice==4:
    gender=input("Person is Adult male or Adult female or Boy or Girl:")
    height=float(input("Enter height: "))
    weight=float(input("Enter weight: "))
    age=float(input("Enter age: "))
    d=bmi(height,weight)
    if gender=="Adult male":
        BFP=1.2*d+0.23*age-16.2
        print("Body fat is",BFP,"%")
    if gender=="Adult female":
        BFP=1.2*d+0.23*age-5.2
        print("Body fat is",BFP,"%")
    if gender=="Boy":
        BFP=1.51*d+0.70*age-2.2
        print("Body fat is",BFP,"%")
    if gender=="Girl":
        BFP=1.51*d+0.70*age-1.4
        print("Body fat is",BFP,"%")
elif choice==0:
    print("Thank you")
    break
else:
    print("Enter a valid choice")
```

Figure 2: NutriFit code, part 2 (TDEE activity levels, Body Fat and Exit)

```

IDLE Shell 3.14.7
File Edit Shell Debug Options Window Help
Python 3.14.7 (tags/v3.14.7:823f032, Aug 5 2026, 10:51:32) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Shivam Singh/Documents/Nutrifit.py =====
=====
NutriFit: An Algorithmic Health & Nutrition Analytics Engine
Analytics Engine - Main Menu
=====
1. BMI(Body Mass Calculator)
2. BMR(Basal Metabolic Rate)
3. TDEE(Total Daily Energy Expenditure)
4. Bodyfat
0. Exit
=====
Enter a choice:1
Enter height in metre:10
Enter weight:20
BMI is 0.2 kg/m² and the person is underweight
=====
1. BMI(Body Mass Calculator)
2. BMR(Basal Metabolic Rate)
3. TDEE(Total Daily Energy Expenditure)
4. Bodyfat
0. Exit
=====
Enter a choice:2
Person is male or female:male
Enter hieght:30
Enter wieght:20
Enter age 2
basal metabolic rate is: 382.5 kilocalories per day
=====
1. BMI(Body Mass Calculator)
2. BMR(Basal Metabolic Rate)
3. TDEE(Total Daily Energy Expenditure)
4. Bodyfat
0. Exit
=====
Enter a choice:0
Thank you
>>>

```

Figure 3: Sample run of NutriFit in IDLE (BMI, BMR and exit)

In this run I chose option 1 (BMI) and then option 2 (BMR), and then typed 0 to exit. The menu was shown again after each calculation, and the program ended with "Thank you".

Menu choice	Input given	Output shown
1 - BMI	height = 10 m, weight = 20 kg	BMI is 0.2 kg/m² and the person is underweight
2 - BMR	male, height = 30, weight = 20, age = 2	basal metabolic rate is: 382.5 kilocalories per day
0 - Exit	0	Thank you (program ends)

10. Testing Approach

I tested the program manually by running it and trying each menu option with normal values, then checking the answer by calculating it myself on a calculator to see if it matched. I did not write automated tests (like pytest) for this version.

- Checked BMI classification by trying values just above and below 18.5, 24.9 and 29.9.
- Checked BMR by running the same height/weight/age once as male and once as female to see the answer changes.
- Checked TDEE by running the same profile through all five activity levels and confirming each gives a different number.
- Checked Body Fat by running each of the four categories once each.

11. Challenges Faced

- The program doesn't check if the input is valid – if the user types letters instead of a number, the program crashes with an error. I didn't handle this in the current version.
- The user has to type the gender/category/activity level exactly as shown (like “male” or “Adult male”) – if they type it slightly differently (extra space, different capitalization) the program just doesn't print anything for that part.
- It took me a while to find the right formulas for BMR and Body Fat Percentage and make sure I typed them correctly in the code.
- Everything is in one file right now, which was easier for me to write and test, but it isn't really split into separate modules/files.

12. Learnings & Key Takeaways

- Writing the repeated formulas (BMI, BMR) as functions instead of typing them again and again made the code shorter and easier to manage.
- Using a while loop with if/elif is a simple way to build a menu-based program.
- I learned that even a simple-looking calculator needs some thought about how to organise the different options (gender, activity level, category) so the right formula is picked each time.
- I also realised how important input validation is after seeing the program crash when I typed something wrong while testing.

13. Future Enhancements

- Add input validation so the program shows an error message instead of crashing when the user types something wrong (using try/except).
- Make the gender/category/activity inputs not case-sensitive, so “Male” and “male” both work.
- Split the code into separate files (like bmi.py, bmr.py, tdee.py, bodyfat.py, main.py) instead of one big file.
- Save the user's results to a file so they can check their progress over time.
- Maybe add a simple GUI later using Tkinter instead of the terminal menu.

14. References

- BMI classification ranges – World Health Organization website.
- BMR formula (Mifflin-St Jeor equation) – referenced from general fitness/health websites.
- Python official documentation – docs.python.org